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## Problem:

- Shared living leads to conflicts over chores, groceries, and expenses
- Existing tools (Splitwise, notes apps, spreadsheets) are fragmented
- No centralized system for managing shared responsibilities
- Lack of automation leads to inefficiency and errors

## Solution:

- NARLA is an AI-driven apartment co-pilot
- Centralizes chores, bills, and inventory management
- Uses predictive AI to automate reminders and tasks
- Improves collaboration and accountability among roommates

## Goals:

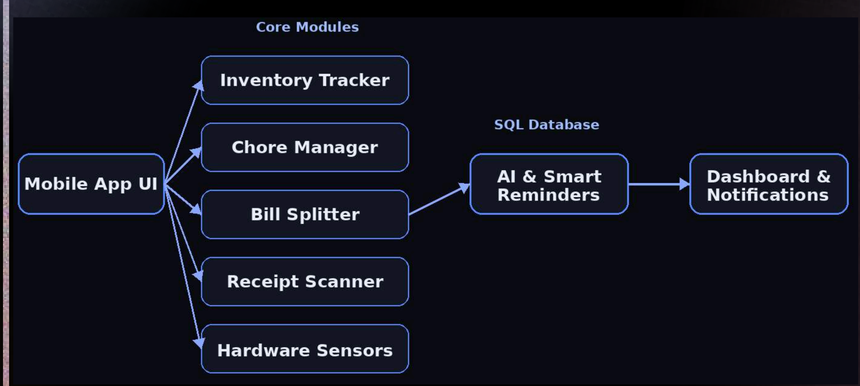
- Automate shared living management
- Reduce conflicts and miscommunication
- Improve organization and transparency
- Provide intelligent, data-driven assistance

LIVE DEMO AVAILABLE!

# narla.

## The Apartment Co-Pilot

## System Architecture:




## Key Features:

- Smart Task & Bill Management
- Tracks chores, expenses, and inventory
- Ensures fair distribution of responsibilities

## AI-Powered Predictions:

- Analyzes usage patterns
- Generates reminders and task suggestions

## OCR Receipt Scanning (Planned):

- Extracts structured data from receipts
- Automates expense tracking

## Inventory Tracking:

- Monitors shared resources
- Predicts restocking needs

## Tech Stack:

- Frontend: React Native
- Backend: Python (Flask / FastAPI)
- Database: PostgreSQL
- AI: Predictive analytics + rule-based models
- Integration: OCR + optional IoT sensors

Flask SQL

## Current Progress:

- Designed normalized PostgreSQL database schema
- Built backend API structure and routing
- Developed frontend navigation (login, dashboard, household views)
- Established frontend-backend communication

## Challenges:

- Multi-user database design without duplication
- Real-time frontend-backend synchronization
- Designing meaningful AI prediction logic
- Balancing software + hardware scope

## Impact:

- Reduces roommate conflicts through transparency
- Improves financial responsibility (bill tracking)
- Encourages sustainable consumption (inventory tracking)
- Supports student productivity and well-being
- Scalable to co-living, families, and assisted living

## Future Work:

- AI prediction improvements
- OCR integration completion
- IoT sensor integration
- Full system deployment and scaling
- Mobile app refinement

## Scan to Explore:

GitHub Repository

Project Demo Video:

Website: